

PLSC 504 – Fall 2026

Class Intro + Review: Likelihoods, Optimization, etc.

August 24, 2026

Class details:

- “Proseminar” in methods
- We’ll be meeting Monday mornings from 09:00-12:00 ET in the Business Building, Room 206
- Texts: Various (mostly articles; see the syllabus)
- All course materials:
<https://github.com/PrisonRodeo/PLSC504-2026-git>
- Methods Preceptor: [Ludmila Ferreira Bandeira](#)
- Software: R...
- Grading: Ten homework assignments (@ 50 points), plus a final project (500 points)
- Contact me: zorn@psu.edu, or text me at +1 (803) 553-4077

A Model:

$$Y \sim N(\mu, \sigma^2)$$

$$\begin{aligned} E(Y) &= \mu \\ \text{Var}(Y) &= \sigma^2 \end{aligned}$$

And some data:

$$\begin{aligned} Y &= 74 \\ &73 \\ &69 \\ &81 \\ &78 \end{aligned}$$

Model implies:

$$\Pr(Y_i = y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(Y_i - \mu)^2}{2\sigma^2} \right]$$

So

$$\Pr(Y_1 = 74) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(74 - \mu)^2}{2\sigma^2} \right]$$

$$\Pr(Y_2 = 73) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(73 - \mu)^2}{2\sigma^2} \right]$$

.

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.

Joint probabilities are:

$$\Pr(A, B | \Pr(A) \perp \Pr(B)) = \Pr(A) \times \Pr(B)$$

So:

$$\Pr(Y_1 = 74, Y_2 = 73) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(74 - \mu)^2}{2\sigma^2}\right] \times \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(73 - \mu)^2}{2\sigma^2}\right]$$

$$\begin{aligned}\Pr(Y_i = y_i \forall i) &\equiv L(Y|\mu, \sigma^2) \\ &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_i - \mu)^2}{2\sigma^2} \right]\end{aligned}$$

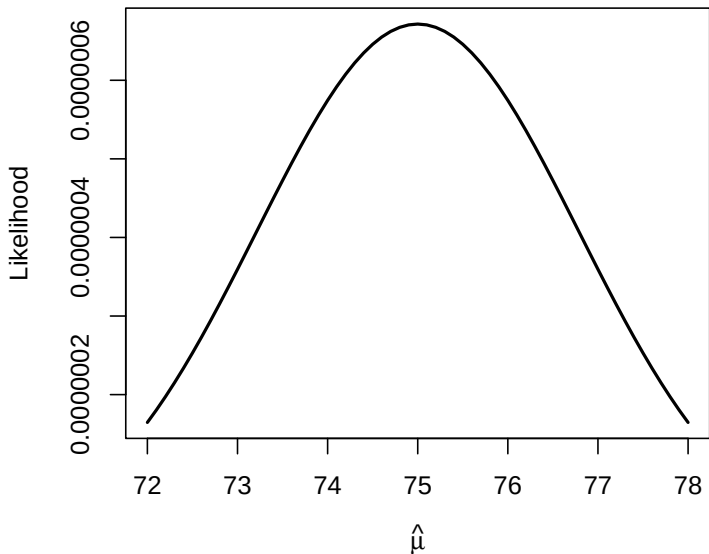
The *likelihood*:

$$L(\hat{\mu}, \hat{\sigma}^2 | Y) \propto \Pr(Y | \hat{\mu}, \hat{\sigma}^2)$$

For $\hat{\mu} = 78$, $\hat{\sigma} = 4$:

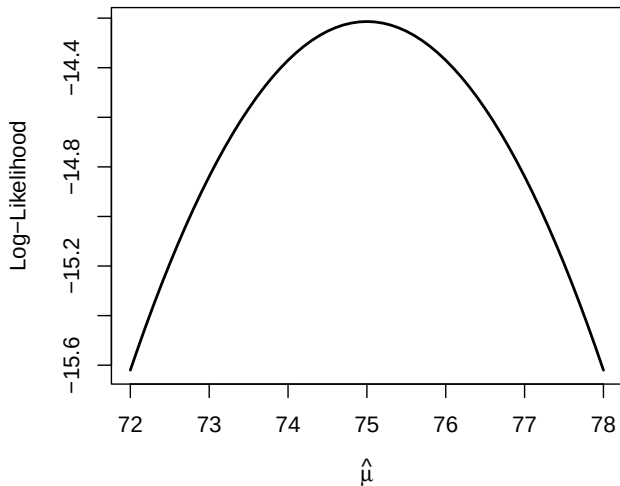
$$\begin{aligned} L &= \frac{1}{\sqrt{2\pi}16} \exp \left[-\frac{(74 - 78)^2}{32} \right] \times \\ &\quad \frac{1}{\sqrt{2\pi}16} \exp \left[-\frac{(73 - 78)^2}{32} \right] \times \\ &\quad \frac{1}{\sqrt{2\pi}16} \exp \left[-\frac{(69 - 78)^2}{32} \right] \times \dots \\ &= \text{some reeeeeally small number...} \end{aligned}$$

What a Likelihood Looks Like



$$\begin{aligned}\ln L(\hat{\mu}, \hat{\sigma}^2 | Y) &= \ln \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(Y_i - \mu)^2}{2\sigma^2} \right] \\ &= \sum_{i=1}^N \ln \left\{ \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(Y_i - \mu)^2}{2\sigma^2} \right] \right\} \\ &= -\frac{N}{2} \ln(2\pi) - \left[\sum_{i=1}^N \frac{1}{2} \ln \sigma^2 + \frac{1}{2\sigma^2} (Y_i - \mu)^2 \right]\end{aligned}$$

What a Log-Likelihood Looks Like



The “Maximum” Part

For $L = f(Y, \theta)$,

- Calculate $\frac{\partial \ln L}{\partial \theta}$,
- Set $\frac{\partial \ln L}{\partial \theta} = 0$, solve for $\hat{\theta}$,
- Calculate $\frac{\partial^2 \ln L}{\partial \theta^2}$,
- Verify $\frac{\partial^2 \ln L}{\partial \theta^2} < 0$.

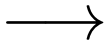
Example: Normal Y

$$\ln L(\hat{\mu}, \hat{\sigma}^2 | Y) = -\frac{N}{2} \ln(2\pi) - \left[\sum_{i=1}^N \frac{1}{2} \ln \sigma^2 + \frac{1}{2\sigma^2} (Y_i - \mu)^2 \right]$$

$$\frac{\partial \ln L}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^N (Y_i - \mu)$$

$$\frac{\partial \ln L}{\partial \sigma^2} = \frac{-N}{2\sigma^2} + \frac{1}{2} \sigma^4 \sum_{i=1}^N (Y_i - \mu)^2$$

Example: Normal Y (continued)



$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N Y_i$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})^2$$

Example: Linear Regression

μ varies linearly in X :

$$\begin{aligned} E(Y) \equiv \mu &= \beta_0 + \beta_1 X_i \\ \text{Var}(Y) &= \sigma^2 \end{aligned}$$

...which implies:

$$L(\beta_0, \beta_1, \sigma^2 | Y) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2} \right]$$

Linear Regression (continued)

Log-likelihood:

$$\begin{aligned}\ln L(\beta_0, \beta_1, \sigma^2 | Y) &= \ln \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(Y_i - \beta_0 - \beta_1 X_i)^2}{2\sigma^2} \right] \\ &= -\frac{N}{2} \ln(2\pi) - \sum_{i=1}^N \left[\frac{1}{2} \ln \sigma^2 + \frac{1}{2\sigma^2} (Y_i - \beta_0 - \beta_1 X_i)^2 \right]\end{aligned}$$

Kernel:

$$- \sum_{i=1}^N \left[\frac{1}{2} \ln \sigma^2 + \frac{1}{2\sigma^2} \underbrace{(Y_i - \beta_0 - \beta_1 X_i)^2}_{\hat{u}_i} \right]$$

$$\Pr(Y) = f(\mathbf{X}, \theta)$$

$$L = \prod_{i=1}^N f(Y_i | \mathbf{X}_i, \theta)$$

$$\ln L = \sum_{i=1}^N \ln f(Y_i | \mathbf{X}_i, \theta)$$

$$\ln L(\hat{\theta} | Y, \mathbf{X}) = \max_{\theta} \{ \ln L(\theta | Y, \mathbf{X}) \}$$

$$\mathbf{g}(\hat{\theta}) = \frac{\partial \ln L(\hat{\theta})}{\partial \hat{\theta}}$$

Taylor series:

$$\frac{\partial \ln L}{\partial \hat{\theta}} \approx \frac{\partial \ln L}{\partial \theta} + \frac{\partial^2 \ln L}{\partial \theta^2}(\hat{\theta} - \theta)$$

$$\begin{aligned}\hat{\theta} - \theta &= \left(-\frac{\partial^2 \ln L}{\partial \theta^2} \right)^{-1} \frac{\partial \ln L}{\partial \theta} \\ &= -\mathbf{H}(\theta)^{-1} \mathbf{g}(\theta)\end{aligned}$$

Need

$$\text{plim}(\hat{\theta} - \theta) = 0$$

So:

- Assume $\mathbf{H}(\theta) \xrightarrow{a} \mathbf{A} < \infty$
- Show $E[\mathbf{g}(\theta)] \rightarrow \mathbf{0}$ as $N \rightarrow \infty$

$$\begin{aligned} \mathbb{E}[\mathbf{g}(\theta)] &= \frac{1}{N} \mathbb{E} \left(\frac{\partial \ln L_1}{\partial \theta} + \frac{\partial \ln L_2}{\partial \theta} + \dots + \frac{\partial \ln L_N}{\partial \theta} \right) \\ &= \frac{1}{N} \left[\mathbb{E} \left(\frac{\partial \ln L_1}{\partial \theta} \right) + \mathbb{E} \left(\frac{\partial \ln L_2}{\partial \theta} \right) + \dots \right] \\ &\stackrel{a}{=} \mathbf{0} \end{aligned}$$

Cramer-Rao say:

$$\text{Var}(\hat{\theta}) \geq \left[-E \left(\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} \right) \right]^{-1}$$

$$\begin{aligned}\text{Var}(\hat{\theta}) &= \text{E}[(\hat{\theta} - \theta)(\hat{\theta} - \theta)'] \\ &= \text{E} \left[\left(-\frac{\partial^2 \ln L}{\partial \theta^2} \right)^{-1} \frac{\partial \ln L}{\partial \theta} \frac{\partial \ln L'}{\partial \theta} \left(-\frac{\partial^2 \ln L}{\partial \theta^2} \right)^{-1} \right]\end{aligned}$$

For MLE:

$$\text{E} \left[\frac{\partial \ln L}{\partial \theta} \frac{\partial \ln L'}{\partial \theta} \right] = \text{E} \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]$$

So,

$$\begin{aligned}\text{Var}(\hat{\theta}) &= \left[-\text{E} \left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) \right]^{-1} \\ &= [\mathbf{I}(\theta)]^{-1}\end{aligned}$$

By LLN:

$$\frac{\hat{\theta} - \theta}{\sqrt{\mathbf{I}(\theta)^{-1}}} \sim N(\mathbf{0}, \mathbf{1})$$

Or:

$$\hat{\theta} \sim N(\theta, \mathbf{I}(\theta)^{-1})$$

For

$$\gamma = h(\theta)$$

$$\hat{\gamma}_{ML} = h(\hat{\theta}_{ML})$$

Suppose

$$\phi^2 = 1/\sigma^2$$

so that

$$Y \sim N(\mu, \phi^2).$$

Then:

$$\ln L(\hat{\mu}, \hat{\phi}^2) = - \left[\sum_{i=1}^N \frac{1}{2} \ln \phi^2 - \frac{1}{2\phi^2} (Y_i - \mu)^2 \right]$$

and:

$$\frac{\partial \ln L}{\partial \phi^2} = \frac{-N}{2\phi^2} + \frac{1}{2}\phi^4 \sum_{i=1}^N (Y_i - \mu)^2$$

and:

$$\begin{aligned} \hat{\phi}^2 &= \frac{N}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \\ &= \frac{1}{\hat{\sigma}^2} \end{aligned}$$

MLEs:

- Maximize $L(\theta|Y, \mathbf{X})$
- Are consistent in N
- Are asymptotically efficient
- Are asymptotically Normal
- Are invariant to (injective) transformations and varying sampling methods

Optimization

The basic problem: find

$$\max_{\hat{\beta} \in \mathbb{R}^k} \ln L(\hat{\beta} | Y, \mathbf{X})$$

The intuition:

- Start with $\hat{\beta}_0$
- Adjust:

$$\hat{\beta}_1 = \hat{\beta}_0 + \mathbf{A}_0$$

- Repeat.

Iterative process:

$$\hat{\beta}_\ell = \hat{\beta}_{\ell-1} + \mathbf{A}_{\ell-1}$$

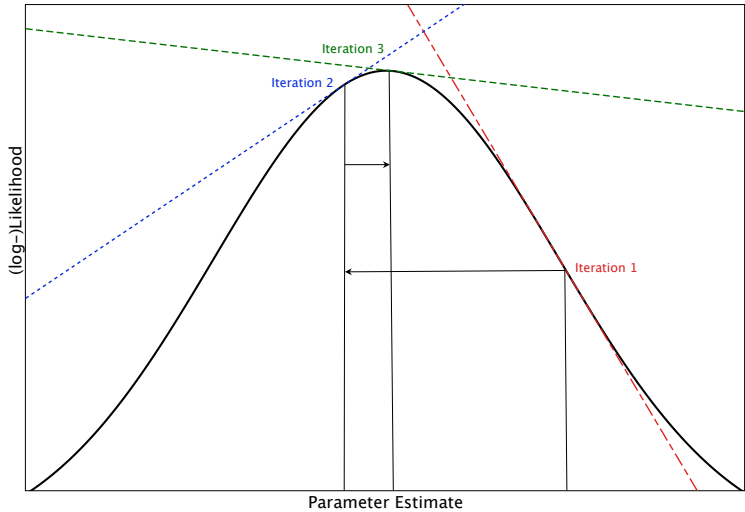
$$\hat{\beta} = \hat{\beta}_\ell \ni \hat{\beta}_\ell - \hat{\beta}_{\ell-1} (\equiv \mathbf{A}_\ell) < \tau$$

How do we adjust?

$$\mathbf{A} = f[\mathbf{g}(\hat{\beta})]$$

Intuition: $\mathbf{g}(\hat{\beta})$ = “directionality” of change

- $\mathbf{g}(\hat{\beta}_k) < 0 \rightarrow A_k < 0$
- $\mathbf{g}(\hat{\beta}_k) > 0 \rightarrow A_k > 0$



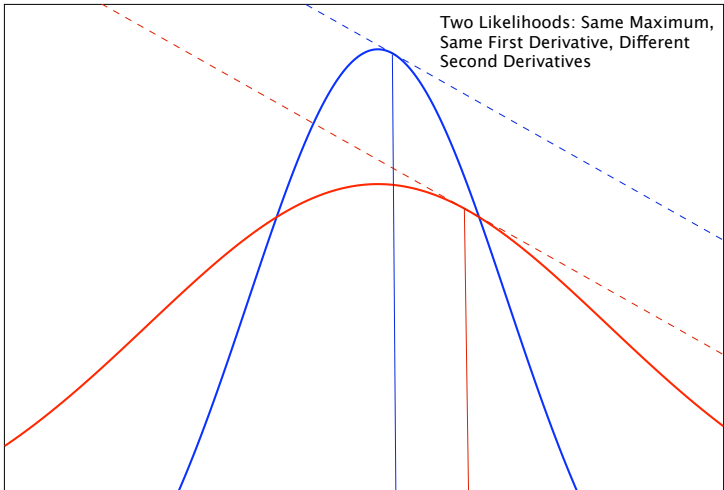
“Steepest Ascent”

Adjustment:

$$\mathbf{A}_\ell = \frac{\partial \ln L}{\partial \hat{\boldsymbol{\beta}}_\ell}$$

At each iteration:

$$\hat{\boldsymbol{\beta}}_\ell = \hat{\boldsymbol{\beta}}_{\ell-1} + \frac{\partial \ln L}{\partial \hat{\boldsymbol{\beta}}_{\ell-1}}$$



“Step size”:

$$\hat{\beta}_\ell = \hat{\beta}_{\ell-1} + \lambda_{\ell-1} \mathbf{\Delta}_{\ell-1}$$

- $\mathbf{\Delta} \rightarrow$ *direction*
- $\lambda \rightarrow$ *amount* (“step size”)

Key: Hessian

$$\mathbf{H}(\hat{\beta}) = \frac{\partial^2 \ln L}{\partial \hat{\beta}^2}$$

How?

$$\begin{aligned}
 \hat{\beta}_\ell &= \hat{\beta}_{\ell-1} - \left(\frac{\partial^2 \ln L}{\partial \hat{\beta}_{\ell-1}^2} \right)^{-1} \frac{\partial \ln L}{\partial \hat{\beta}_{\ell-1}} \\
 &= \hat{\beta}_{\ell-1} - \mathbf{H}(\hat{\beta}_{\ell-1})^{-1} \mathbf{g}(\hat{\beta}_{\ell-1})
 \end{aligned}
 \tag{1}$$

Sidebar: Newton-Raphson, re-revealed

Taylor series, anyone?

$$f(X) \approx f(a) + f'(a)(x - a)$$

Here,

$$\frac{\partial \ln L}{\partial \hat{\beta}_\ell} \approx \frac{\partial \ln L}{\partial \hat{\beta}_{\ell-1}} + \frac{\partial^2 \ln L}{\partial \hat{\beta}_{\ell-1}^2}(\hat{\beta}_\ell - \hat{\beta}_{\ell-1})$$

What we really want...

$$\frac{\partial \ln L}{\partial \hat{\beta}_\ell} = \mathbf{0}$$

So:

$$\mathbf{0} \approx \frac{\partial \ln L}{\partial \hat{\beta}_{\ell-1}} + \frac{\partial^2 \ln L}{\partial \hat{\beta}_{\ell-1}^2} (\hat{\beta}_\ell - \hat{\beta}_{\ell-1})$$

$$\begin{aligned} \hat{\beta}_\ell &\approx \hat{\beta}_{\ell-1} - \left(\frac{\partial^2 \ln L}{\partial \hat{\beta}_{\ell-1}^2} \right)^{-1} \frac{\partial \ln L}{\partial \hat{\beta}_{\ell-1}} \\ &\approx \hat{\beta}_{\ell-1} - \mathbf{H}(\hat{\beta}_{\ell-1})^{-1} \mathbf{g}(\hat{\beta}_{\ell-1}) \end{aligned}$$

Newton-Raphson requires $\mathbf{H}(\hat{\beta})^{-1} \rightarrow$ *calculates* $\mathbf{H}(\hat{\beta})^{-1}$ at every iteration. This can make it somewhat slow / computationally demanding.

Modified Marquardt:

- Used when $\mathbf{H}(\hat{\beta})$ isn't invertable
- Adds a constant $\mathbf{C}$ to $\text{diag}[\mathbf{H}(\hat{\beta})]$
- Variants: Add $\mathbf{C}(h_k)$

"Method of Scoring" (due to Fisher) uses:

$$\begin{aligned}\hat{\beta}_\ell &= \hat{\beta}_{\ell-1} - \left[\mathbb{E} \left(\frac{\partial^2 \ln L}{\partial \hat{\beta}_{\ell-1}^2} \right)^{-1} \right] \frac{\partial \ln L}{\partial \hat{\beta}_{\ell-1}} \\ &= \hat{\beta}_{\ell-1} - \{ \mathbb{E}[\mathbf{H}(\hat{\beta}_{\ell-1})] \}^{-1} \mathbf{g}(\hat{\beta}_{\ell-1})\end{aligned}$$

Berndt, Hall², and Hausman (“BHHH”)

Uses:

$$\hat{\beta}_{\ell} = \hat{\beta}_{\ell-1} - \left(\sum_{i=1}^N \frac{\partial \ln L}{\partial \hat{\beta}_{\ell-1}} \frac{\partial \ln L'}{\partial \hat{\beta}_{\ell-1}} \right)^{-1} \frac{\partial \ln L}{\partial \hat{\beta}_{\ell-1}}$$

Advantages:

- (Relatively) very easy to compute
- Reasonably accurate...

- Davidson-Fletcher-Powell (“DFP”)
- Broyden et al. (“BFGS”)
- They are:
 - Faster / more efficient
 - Comparatively bad at getting $-\left(\mathbf{H}(\hat{\beta})\right)^{-1}$

Summary: Optimization & Inference

Method	"Step size" (∂^2) matrix	Variance-Covariance Estimate
Newton	Inverse of the observed second derivative (Hessian)	Inverse of the negative Hessian
Scoring	Inverse of the expected value of the Hessian (information matrix)	Inverse of the negative information matrix
BHHH	Outer product approximation of the information matrix	Inverse of the outer product approximation

Lots of optimizers:

- `maxLik` package: options for Newton-Raphson, BHHH, BFGS, others
- `optim` (in stats) – quasi-Newton, plus others
- `nlm` (in stats) – nonlinear minimization “using a Newton-type algorithm”
- `newton` (in Bhat) – Newton-Raphson solver
- `solveLP` (in linprog) – linear programming optimizer

- *Must* provide log-likelihood function
- Can provide $\mathbf{H}(\hat{\beta})$, $\mathbf{g}(\hat{\beta})$, both, or neither
- Choose optimizer (Newton, BHHH, BFGS, etc.)
- Returns an object of class `maxLik`

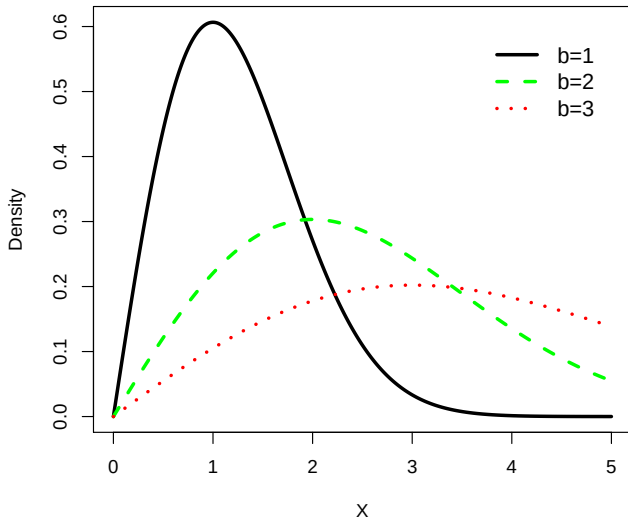
Rayleigh distribution density:

$$\Pr(X = x) = \frac{x}{b^2} \exp \left[\frac{-x^2}{2b^2} \right], \quad b > 0$$

Other traits:

- Support $\in [0, \infty)$
- $E(X) = b\sqrt{\frac{\pi}{2}}$
- Mode = b
- $Var(X) = \frac{4-\pi}{2} b^2$

Rayleigh Densities



We can generate a Rayleigh-distributed random variable X with parameter b via inverse transform sampling, as:

$$X = b\sqrt{-2\ln(1 - U)}$$

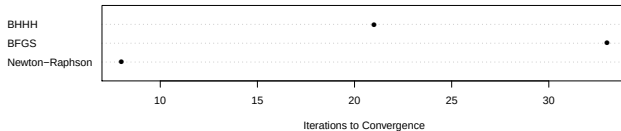
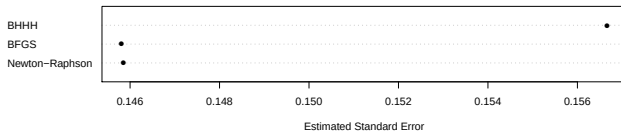
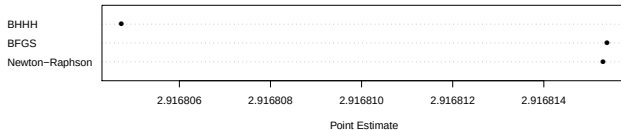
where $U \in \text{Uniform}[0, 1]$. So, for (e.g.) $b = 3$:

```
> library(maxLik,distr)
> set.seed(7222009)
> U<-runif(100)
> rayleigh<-3*sqrt(-2*log(1-U)) # b = 3
> loglike <- function(param) {
+   b <- param[1]
+   ll <- (log(x)-log(b^2)) + ((-x^2)/(2*b^2))
+   ll
+ }
```

R : What We Like To See

```
> x<-rayleigh
> hats <- maxLik(loglike, start=c(1))
> summary(hats)
-----
Maximum Likelihood estimation
Newton-Raphson maximisation, 8 iterations
Return code 2: successive function values within tolerance limit
Log-Likelihood: -195.7921
1 free parameters
Estimates:
      Estimate Std. error t value Pr(> t)
[1,]    2.9168    0.1459     20 <2e-16 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1  1
-----
```

Comparing Optimizers



R : What We *Don't* Like To See

```
> Y<-c(0,0,0,0,0,1,1,1,1,1)
> X<-c(0,1,0,1,0,1,1,1,1,1)
> logL <- function(param) {
+   b0<-param[1]
+   b1<-param[2]
+   ll<-Y*log(exp(b0+b1*X)/(1+exp(b0+b1*X))) +
+       (1-Y)*log(1-(exp(b0+b1*X)/(1+exp(b0+b1*X))))
+   ll
+ }
```


R : What We *Don't* Like To See

```
> Bhat<-maxLik(logL,start=c(0,0))  
> summary.maxLik(Bhat)
```

```
-----  
Maximum Likelihood estimation  
Newton-Raphson maximisation, 9 iterations  
Return code 1: gradient close to zero  
Log-Likelihood: -4.187887  
2 free parameters  
Estimates:
```

	Estimate	Std. error	t value	Pr(> t)
[1,]	-104.3	Inf	0	1
[2,]	105.2	Inf	0	1

```
-----
```

- Potential Problems
- Likely Causes
- Tips

Enemy # 1: Noninvertable $\mathbf{H}(\hat{\beta})$

- “Non-concavity,” “non-invertability,” etc.
- (Some part of) the likelihood is “flat”
- Why? (Bob Dole...)

Identification

- Possible due to functional form alone...
- “Fragile”
- Manifestation: parameter instability

Poor Conditioning

- Numerical issues
- Potentially:
 - Collinearity
 - Other weirdnesses (nonlinearities)

- Misspecification.
- Missing data
- Variable scaling
- Typical $\Pr(Y)$

- T-h-i-n-k!
- Know thy data
- Keep an eye on your iteration logs...
- Don't overreach